(SHORT)

Microscope info:

Image		2026-02-03-LSM800_63x_Coreg
image's creation	date	
	method used	
Actual image depth		16
Microscope type		WideField
Objective	NA	1.4
	im. refractive index	1.518

Channel(s)	Wavelengths		Saturation	sampling (X,Y,Z)		
	Ex. (nm)	Em. (nm)		Nyquist (µm)	Found (µm)	Nyquist/found ratio
Channel 0		668.0	none	0.119x0.119x 0.359	0.198x0.198x 0.2	1.7, 1.7, 0.6
Channel 1		603.0	none	0.108x0.108x 0.324		1.8, 1.8, 0.6
Channel 2		517.0	none	0.092x0.092x 0.278		2.1, 2.1, 0.7
Channel 3		465.0	none	0.083x0.083x 0.25		2.4, 2.4, 0.8

Warnings:

(No saturated pixels detected).

The highlighted undersampled channels may alter the result interpretation.
(The bead size is appropriate for this coalignment analysis).

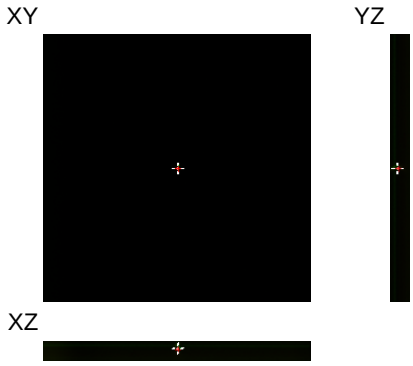
Ratios table:

	Channel 0	Channel 1	Channel 2	Channel 3
Channel 0		0.604	1.077	41.73
Channel 1	0.604		0.479	41.545
Channel 2	1.077	0.479		41.322
Channel 3	41.73	41.545	41.322	
X, Y, Z theoretical resolutions (in µm)	0.243, 0.243 & 0.916	0.22, 0.22 & 0.827	0.188, 0.188 & 0.709	0.169, 0.169 & 0.637
Bead centres'coord.(X,Y & Z in µm)	50.905, 50.594 & 2.444	50.876, 50.622 & 2.975	50.835, 50.658 & 3.363	41.146, 53.296 & 5.321
Bead quality (SB Ratio)	102.6	107.0	100.4	7.3
Title	C1-2026-02-03-LSM800_63x_Coreg.czi	C2-2026-02-03-LSM800_63x_Coreg.czi	C3-2026-02-03-LSM800_63x_Coreg.czi	C4-2026-02-03-LSM800_63x_Coreg.czi

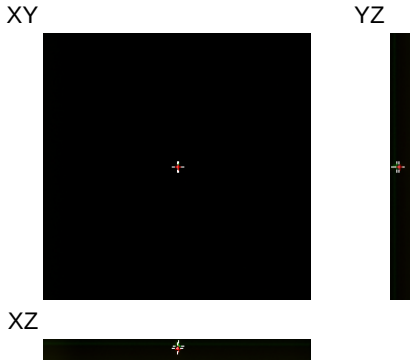
Ratio of the mesured calibrated distance to the reference distance (as calulated using the longest wavelength).
Green: within specifications, red: outside specifications (ie. ratio above 1.0)

Profile view:

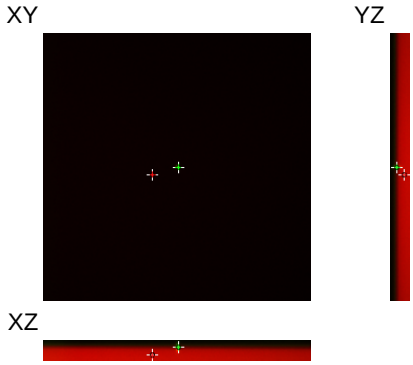
Channel 0 (Em. Wavelength 668.0 nm) vs channel 1 (Em. Wavelength 603.0 nm)



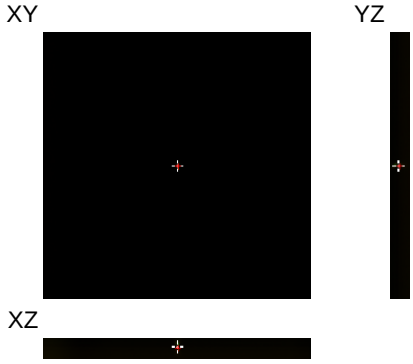
Channel 0 (Em. Wavelength 668.0 nm) vs channel 2 (Em. Wavelength 517.0 nm)



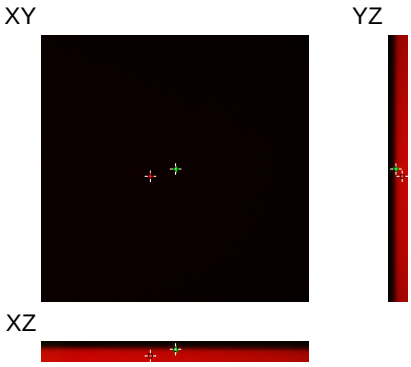
Channel 0 (Em. Wavelength 668.0 nm) vs channel 3 (Em. Wavelength 465.0 nm)



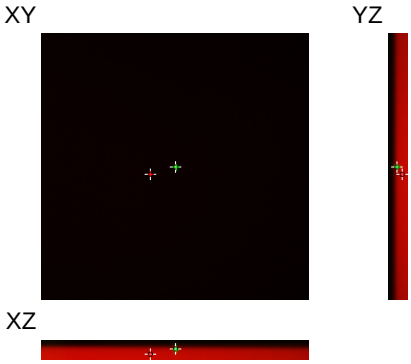
Channel 1 (Em. Wavelength 603.0 nm) vs channel 2 (Em. Wavelength 517.0 nm)



Channel 1 (Em. Wavelength 603.0 nm) vs channel 3 (Em. Wavelength 465.0 nm)



Channel 2 (Em. Wavelength 517.0 nm) vs channel 3 (Em. Wavelength 465.0 nm)



Analysis parameters

Tool & Operator	Tool	Co-registration
	Versions	MetroloJ_QC v1.3.1.2, ImageJ v2.16.0/1.54p, Java v11.0.31, OS Windows 10
	Operator & date	, 23 April 2026, 15:06
data	result folder	G:\QC\\
	Type of saved data	.pdf
	Input data bit depth	16
Dimension order		XY-(C)Z
Discard saturated samples		false
Beads	Bead detection threshold	Otsu
	Center detection method	Centroid
	Discard bead if more than one particle are thresholded	false
	Background annulus thickness in µm	1.0
	Background annulus distance to bead edges in µm	0.5
	Multiple beads in image	false
Tolerance	Applied in this report	true
	Ratio valid if below	1.0

Formulas used:

The reference distance is calculated as follows:

$$r_{ref} = \frac{res_x^o * res_y^o * res_z^o}{\sqrt{(res_y^o * res_z^o * cos\Phi * sin\Theta)^2 + (res_x^o * res_z^o * sin\Phi * sin\Theta)^2 + (res_x^o * res_y^o * cos\Theta)^2}}$$

$$\Phi = \arccos \frac{x_B - x_A}{\sqrt{(x_B - x_{A)}{}^2 + (y_B - y_{A)}{}^2}}$$

$$\Theta = \arccos \frac{z_B - z_A}{\sqrt{(x_B - x_{A)}{}^2 + (y_B - y_{A)}{}^2 + (z_B - z_{A)}{}^2}}$$

$res_{x,y}^o = \frac{0.51*\lambda_{em}}{NA}$ $\qquad$ $res_z^o = \frac{\lambda_{em}}{n-\sqrt{n^2 - NA^2}}$

x_A, y_A, z_A and x_B, y_B, z_B are the bead coordinates in channel A and B respectively, NA: numerical aperture, λ_{em}: emission wavelength, n: refractive index of the lens immersion & mounting media.

Lateral ($res_{x,y}^o$) and axial (res_z^o) theoretical resolution values used for widefield microscopes are calculated as defined in Wilhelm, S. Confocal Laser Scanning Microscopy, 2011.

Compliance with the Shannon-Nyquist criterion uses the following formulas for Shannon-Nyquist distances calculation:

$\alpha = \arcsin(\frac{\mathrm{NA}}{\mathrm{n}})$

$\Delta_{x,y} = \frac{\lambda_{em}}{4.NA}$ $\qquad$ $\Delta_z = \frac{\lambda_{em}}{2.n.\,(1-cos\,(\alpha))}$